Integration: An emerging service

So now, the cloud is raining resources. All we have to do is catch whatever we need in a tub, and start using it (and of course, pay for it too). But who does that? Cloud integration is no mean task, as it involves integrating not just many cloud-based resources, but also integrating legacy non-cloud resources with the cloud-based ones!

Cloud-computing pioneers such as Amazon EC2 and Force.com have their own application platforms, in addition to the computing resources, so developers can easily build their applications on the platform and get going in no time. But that is just one of the simpler options. You can also do a complete custom build, by selecting disparate resources and putting these together, not to forget integrating the company's pre-cloud resources.

There are many integration tools available, including software-based, appliance-based, and cloud-delivered ones. Developers can choose from amongst these integration tools to build their solutions. Cloud integration is emerging as a specialised genre of development and service. It is also a popular start-up idea!

A rather large number of such open source tools are now available, right from platforms and development tools, to management dashboards and automated migration tools for applications. Existing open source platforms are also fast adapting to the needs of cloud computing and include features such as intelligent workload management and cloud-enabled scalability considerations, to help massive horizontal scalability at all the layers of the technology stack. Ubuntu, Red Hat, and almost every other open source platform now has a stable cloud offering.

George Paul, executive vice president, HCL Infosystems, quickly justifies our point with some examples: "An open source software-infrastructure project called Eucalyptus imitates the experience of using Amazon's EC2, but allows users to run programs on their own resources. The University of Chicago's Nimbus is another open source cloud-computing project that is widely recognised as having pioneered the field. Today, customers have a large choice of open source applications for the cloud, including Red Hat, Traffic Server, Puppet, Zoho, Cloudera, Enomaly and Joyent."

He goes on to explain that apart from new tools, existing open source offerings are being made cloud-enabled through standardisation. "A common standard called the Application Packaging Standard (APS), an open standard with all specifications, has been introduced; it helps in making applications multi-tenant. It consists of over 250 applications, and will continue to grow in the future as well," he says. APS helps to standardise the packaging, automate the provisioning and management, and to integrate with other hosted services.

HCL's O'zone, a cloud-enabled services suite, combines a variety of open source solutions, including the Proxmox open source virtualisation platform, Red Hat Enterprise Virtualisation, and open source enterprise resource planning, customer relationship management and content management systems.

Open source for the cloud

Virtualisation of infrastructure is at the heart of cloud computing. Open source options, such as the Kernel-based Virtual Machine (KVM) for Linux and the Xen hypervisor, are very competent. Other open source, Linux-native tools like Hadoop, Cassandra, HipHop, CouchDB and Btrfs also assist one in building a first-class data centre, very cost-effectively. These could be seen as the starting point for the coming wave of enterprise-scale open source adoption on the cloud.

Software platforms such as Eucalyptus, which enable the implementation of private and hybrid clouds, are also becoming very popular. Eucalyptus is a modular platform that is capable of working with a variety of interfaces, including Amazon's EC2 and Simple Storage Service (S3) services. Eucalyptus works with various distros, including Red Hat Enterprise Linux (RHEL), CentOS, SUSE Linux Enterprise Server (SLES), OpenSUSE, Debian and Fedora. It can also host MS Windows images. It is capable of working with many virtualisation technologies such as VMware, Xen and KVM hypervisors, in order to implement the abstraction demanded by a cloud environment.

Solutions such as Cloudera build on the capabilities of popular open source options like Hadoop, to meet an enterprise's cloud-computing needs. The Cloudera data management platform incorporates the Hadoop Distributed File System (HDFS), Hadoop MapReduce, Hive, Pig, HBase, Sqoop, Flume, Oozie, Zookeeper and Hue, and is available free under an Apache licence. The enterprise package includes support, tools and training.

Joyent's SmartPlatform is an open source, server-side JavaScript-based framework for developing and delivering real-time, asynchronous Web applications to the cloud. It is basically a platform-as-a-service. While hosting is free at the moment, it might become a paid service once SmartPlatform graduates from the beta to a stable release. People are betting big on Joyent's offering, because of the ubiquity of Javascript.

Enomaly's open source cloud-management and provisioning software, the Elastic Computing Platform or ECP, allows an enterprise to create a private cloud inside its own data centres. It can also link the private cloud to a public one, as and when the company suddenly needs more computing resources. The tool-kit can also be used by service providers and telecom firms to quickly set up and deliver infrastructure-as-a-service (IaaS) cloud-computing services to customers. ECP includes security and compliance features, and Enomaly also offers service and support to licence holders, in a model very similar to that followed by companies like Red Hat. RightScale,